MAYNOOTH UNIVERSITY

DEPT. ELECTRONIC ENGINEERING

EE695

COVERAGE AND CAPACITY ANALYSIS OF LOW EARTH ORBIT SATELLITES

A PROJECT REPORT SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS OF THE MAYNOOTH UNIVERSITY FOR THE DEGREE OF MASTERS OF ENGINEERING

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Overview of LEO constellations (Starlink, OneWeb, Kuiper)
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A. Data Inputs

The main data input to the propagation of satellites in this project is Two-Line Element sets (TLEs). TLEs are a common publicly accessible format which describes Earth orbiting satellite positions. The primary source of TLE data is SpaceTrack [**?**]. Although SpaceTrack is the primary source many projects such as this project and other research rely on on sources such as CelesTrak which takes the data from SpaceTrack and organizes it into specific groupings [**?**]. Some examples of these groupings can be seen listed in Listing **??**

- **Starlink**
- **ISS**
- **Earth observation satellites**
- **Weather satellites**
- **Communications satellites**
- **Active vs. debris groupings**

While other sites exist such as N2Y0 and AMSAT , CelesTrak remains one of the most reliable and widely used platforms due to consistent updates which are synchronized with SpaceTrack and categories which help simplify data access. Celetrak updates the majority of its TLE files approximately every two hours. Celestrak's starlink data can expect multiple updates per day, making it suitable for this projects application [**?**].

B. Two Line Element Data

TLEs are a standardized format which are used to describe the orbital properties of any Earth orbiting satellites. A TLE is a representation of a satellite's orbit and a specific moment in time, this is known as the epoch. Because the TLE is simple and lightweight it is widely support and the most common way to distribute orbit data for tracking and prediction.

A TLE consists of three lines, line 0, which is optional contains the satellite name, line 1 contains information relating to satellite identification and basic orbital parameter and line 2 describes the remaining orbital elements such as the inclination angle. This can be seen in further detail in **??**. An example of TLE data can also be seen in **??**

```
STARLINK-5694
1 55481U 23015AJ  26041.24424508  .00002920  00000+0  23114-3 0  9992
2 55481  42.9978 128.2652 0001315 248.0292 112.0411 15.02536911166474
```

Listing 1: Two-Line Element (TLE) for STARLINK-5694

Table I: Summary of Key Parameters in Two-Line Element Set

Field	**Value**	**Meaning**
Satellite Name	STARLINK-5694	Human-readable satellite identifier
Catalog Number	55481	Unique NORAD satellite ID
Classification	U	Unclassified satellite
Launch Year	2023	Part of the International Designator (23)
Launch Number	015	15th launch of the year 2023
Launch Piece	AJ	Specific object from that launch
Epoch (Year)	2026	TLE epoch year (26)
Epoch (Day)	41.24424508	Day of year 2026 when the TLE is valid
1st Derivative of Mean Motion	0.00002920	Rate of orbital decay (drag)
2nd Derivative of Mean Motion	0.00000	Rarely used; typically zero
BSTAR Drag Term	2.3114e-3	Atmospheric drag coefficient
Ephemeris Type	0	Always zero for SGP4
Element Set Number	999	TLE version counter
Inclination (deg)	42.9978	Tilt of orbit relative to Earth's equator
RAAN (deg)	128.2652	Longitude where orbit crosses equator going north
Eccentricity	0.0001315	Orbit shape (nearly circular)
Argument of Perigee (deg)	248.0292	Orientation of orbit's closest point
Mean Anomaly (deg)	112.0411	Satellite's position in orbit at epoch
Mean Motion (rev/day)	15.02536911	Orbits per day (≈95.8 min period)
Revolution Number	166474	Number of completed orbits since launch

C. Orbit Propagation

Why SGP4

How SGP4 works at a high level

The Simplified General Perturbations model 4 (SGP4) [**?**]

D. Coordinate Transformations

Coverage estimation first requires the satellite and ground location to be expressed in a common frame/coordinate system at a given epoch. Satellites which are propagated using SGP4 are expressed in True Equator, Mean Equinox frame (TEME), while ground sites are typically given in geodetic coordinates (latitude and longitude). The geodetic frame is tied to the Earth and rotates with it, unlike TEME which is fixed to the stars. It is therefore necessary to convert both of these into a common frame so that necessary calculations for visibility and choosing an optimal satellite can be carried out. This was done by converting the ground site and satellite to an Earth Centered Earth Fixed (ECEF) frame [**?**] [**?**].

1) TEME: TEME is the native coordinate output of SPG4 for a satellite's position is generated from TLE data. It uses Earth's instantaneous (true) equator at an epoch combined with the mean equinox which helps to average out potential short term wobbles in the Earth's rotation. TEME is non-rotating in

respective to Earth's surface. This makes it so you cannot directly compare TEME positions to geodetic locations and TEME must first be converted into ECEF [**?**].

2) ECEF: The ECEF frame is a rotating cartesian coordinate system whose origin is at the Earth's center of mass. Unlike TEME, ECEF rotates with the Earth, meaning a fixed point on the surface will stay constant with a given set of coordinates over a period of time. The axes are defined relative to the Earth's geometry. The Z-axis alings with the rotation axis toward the north pole, the X-axis passes through the intersection between the equator and Greenwich meridian, and the Y-axis complete the right handed system at 90° east [**?**]. This can be visualized in figure **??**.

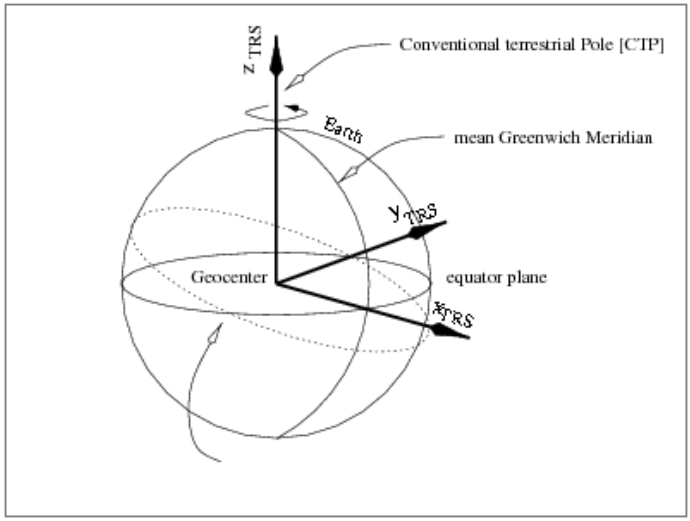

Figure 1: Diagram displaying ECEF frame [**?**]

3) TEME → ECEF: As discussed in section **??**, to compare a satellite from SGP4 with a ground point, the TEME vector must be rotated by Earth's sidereal rotation. An example of this rotation can be seen in (**??**).

$$\begin{bmatrix} x_{\text{ECEF}} \\ y_{\text{ECEF}} \\ z_{\text{ECEF}} \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{R_3(+\theta)} \begin{bmatrix} x_{\text{TEME}} \\ y_{\text{TEME}} \\ z_{\text{TEME}} \end{bmatrix} \tag{1}$$

This aligns the satellite position with a Earth fixed rotating frame.

4) Geodetic → ECEF: To compare a satellite with an observer on Earth, the observer's latitude, longitude and altitude must be expressed as cartesian X, Y, Z coordinates in the ECEF frame. As the Earth is not a perfect sphere it is approximated as a WSG-84 ellipsoid. This is defined by it's semi major axis(the equatorial size), the flattening(how much the poles are compressed) and the first eccentricity (a measure of how non spherical the ellipsoid is). This can be seen in (**??**) [**?**].

$$e^2 = 2f - f^2. \tag{2}$$

As the Earth bulges at the equator, the distance from the center to the surface will change with latitude. This can be handled using a prime-vertical radius of curvature as seen in (**??**) [**?**].

$$N(\phi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \phi}} \tag{3}$$

Once $N(\phi)$ has been found the observer location can be described in ECEF as seen in (**??**) [**?**].

$$
\begin{aligned}
X &= (N(\phi) + h)\ \cos\phi\ \cos\lambda, \\
Y &= (N(\phi) + h)\ \cos\phi\ \sin\lambda, \\
Z &= \big(N(\phi)(1 - e^2) + h\big)\ \sin\phi.
\end{aligned} \tag{4}
$$

These equations convert the geodetic position (ϕ, λ, h) and the satellite's equation into the ECEF frame. This allows for all later geometry such as line of sight checks to be carried out.

E. Satellite Visibility Model

Determining whether a satellite can provide coverage requires evaluating the geometric line of sight(LOS) between the observer location and the satellite. Once both of these positions are expressed in ECEF, visibility can be computed using LOS vector analysis, local horizon projection and elevation angle masking.

The first step is to compute the vector from the observer to the satellite,

$$d = r_{sat} - r_{obs}. \tag{5}$$

This vector expresses the direction of the satellite relative to the observers location. IF the satellite lies below the horizon it cannot be deemed visible.

To know if the satellite appears in the sky from an observers perspective, the LOS vector is transformed into an East-North-Up (ENU) frame. This rotates the ECEF coordinates to be aligned with the observer's.

- East (E) - horizontal axis pointing east
- North (N) - horizontal axis pointing north
- Up (U) - vertical axis pointing outward from Earth.

This transformation allows for the elevation angle to be computed [**?**]. It cal also be visualized in figure.

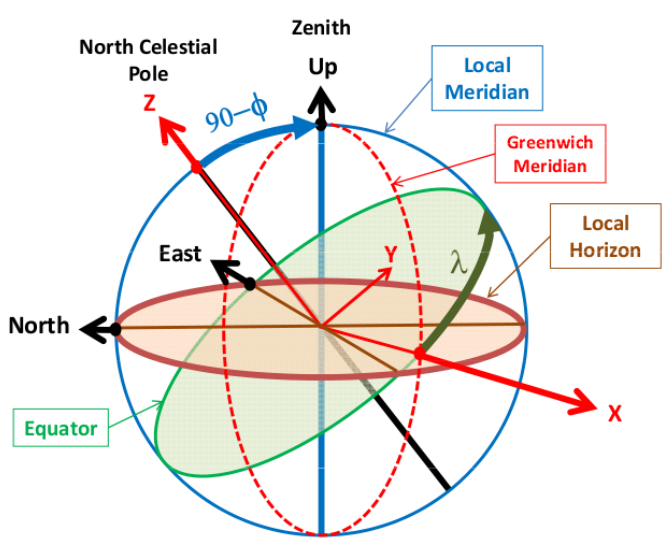

Figure 2: Transformation between ENU and ECEF [**?**]

The elevation angle then describes how high above the horizon a satellite appears as seen in (**??**)

$$\theta_{el} = arctan\left(\frac{u}{\sqrt{e^2 + n^2}}\right) \tag{6}$$

The result can be interpreted as follows:

$$\theta_{\text{el}} > 0^\circ \ : \ \text{satellite is above the geometric horizon,}$$

$$\theta_{\text{el}} = 0^\circ \ : \ \text{exactly on the horizon,}$$

$$\theta_{\text{el}} < 0^\circ \ : \ \text{below the horizon (not visible).}$$

However, in reality, satellites close to the horizon may suffer from increased attenuation, obstruction from terrain and buildings, and doppler shifts. For this reason, a 10° elevation mask is applied. Only satellites which exceed this angle are deemed visible [**?**]. This can be seen in figure **??**, where β is the masking angle.

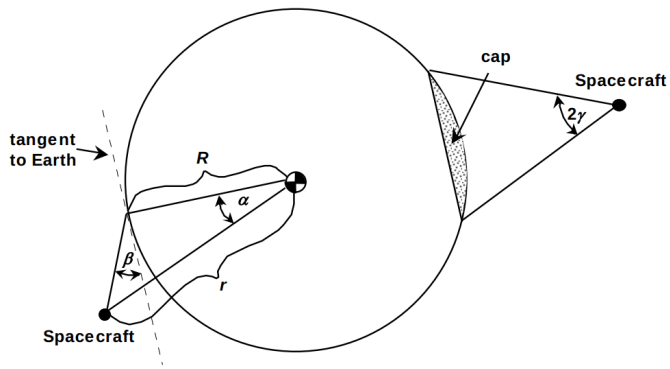

Figure 3: Masking angle [**?**]

F. beam-width Model

To estimate when a satellite can provide service to a ground location, just the LOS is not sufficient. Even if the satellite can be visible from an observer location, the end user must also fall into the satellite's communication beam footprint. This project uses a simplified nadir-pointing beam-width model to determine if the satellites cap, which can be seen in figure **??**, covers the observe's position.

Modern LEO Satellite's used for communications such as Starlink, use phased-array antennas that steer beams electronically. However, for the high level coverage analysis in this project, the beam is approximated as a fixed beam directed toward the nadir (the point directly below the satellite).

The beam can be characterized by its half power beam-width which defines the angular radius of the cap

$$\theta_{beam}. \tag{7}$$

This angle represents how far off the nadir a user can be while still remaining within the satellites service region. LEo satellites can range from narrow high gain spot beams with a width of 1° to 5° for mmWave communications to wider beams which can range from 10° to 100° for more broad coverage [**?**].

This is the basis of the second visibility filter which will be implemented, which works with the LOS filter. Even if a satellite is above the horizon and can be seen from an observer location, it cannot provide coverage unless the user is also within the cap.

G. Coverage Window Detection

Once a satellite and it's footprint have been evaluated the next step is to determine the the coverage window. This is the interval of time which a specific satellite can be used to provide coverage for a user.

As LEO satellites move rapidly across the sky, these windows are short and occur multiple times a day, the detection of these windows is import to help identify key metrics such as the optimal satellite at a given time.

To determine the visibility window, the system must evaluate the satellite positons at discrete time steps. For each time instant, the satellites positions are propagated with SGP4, the satellites elevatin angle relative to the observer is computed, an elevation mask is applied and the nadir angle is compared to the beam-width limit.

A coverage window comes from the transition between in-coverage to not in-coverage state. A coverage window will begin when, from the user's position, the state changes from no coverage to having coverage, and the window will end when there is no satellite coverage.

For each satellite which provides coverage at any point during the simulation, the system will output a start time of coverage, an end time of coverage, the duration of the coverage, and if there is more than one satellite able to provide coverage, which is the optimal satellite.

H. Coverage Analysis

Link budget approximations
Capacity estimation model

IV. Implementation

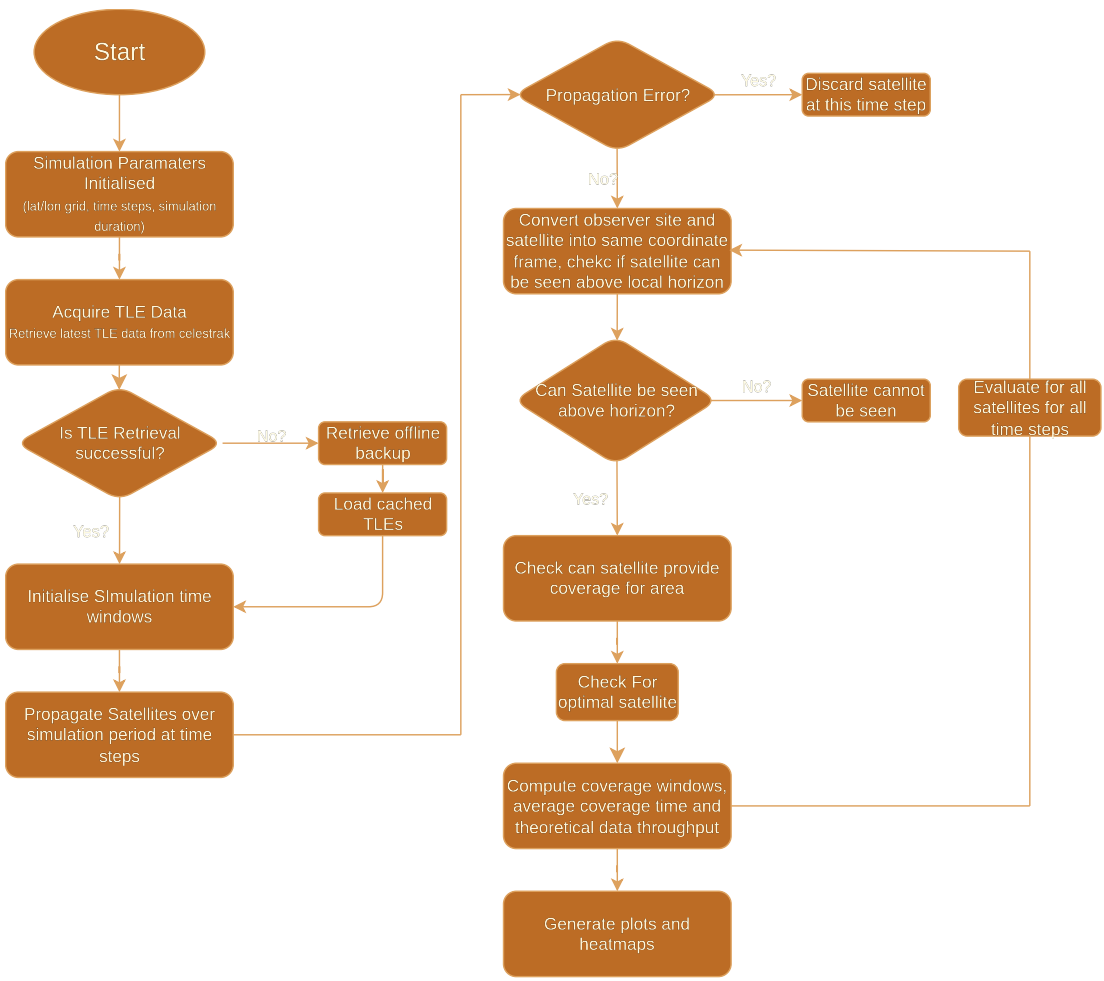

Figure 4: Implementation Logic

The implementation of this project consists of a Python based system which retrieves live orbital data for LEO satellites using Celestrak and then propagates each orbit using the SGP4 model. Using these propagations, geometric visibility checks for specific locations are calculated. The system is designed to support coverage and capacity calculations by determining which satellites are optimal for downlink access at a given period in time.

A. Data acquisition

The first stage of implementation is acquiring up ot date orbital Parameters for the Starlink LEO constellation. TLE sets are retrieved from Celestrak. This data provides the perturbations required to propagate the orbits using SGP4. The TLE data used in this system are taken directly from Celestrak [**?**]. Downloading this data produces a text stream which has sets of three lines of text blocks, satellite name, followed by line 1 and line 2 of the TLE. The implementation of this system parses these entries for each satellite.

Using live TLEs ensures that the system accurately represents the dynamic nature of large LEO constellations, which orbital parameters may change drastically and regularly due to any number of factors such as manual maneuvers.

B. Orbit propagation

Once the TLEs have been retrieved and parsed in a suitable manner, each satellite's position and velocity are computed using the SGP4 algorithm. In this system each satellites state vector can be generated as seen in Listing **??**.

Listing 1: SGP4 propagation example

```
satellite = Satrec.twoline2rv(line1, line2)
e, r, v = satellite.sgp4(jd, fr)
```

The outputs of the **??** are as follows:

- **r**: The satellite position vector (x, y, z) expressed in the True Equator Mean Equinox (TEME) reference frame.
- **v**: The satellite velocity vector in the TEME coordinate frame.
- **e**: An error flag returned by the propagator, where a value of zero indicates successful propagation and non-zero values correspond to specific SGP4 error conditions.

The Julian Date (jd, fr) is computed using the current UTC timestamp to ensure a precise propagation epoch. Any satellite with a non zero error flags are discarded to avoid propagating any invalid orbital data.

C. Coordinate Transformations

Satellite positions produced by SGP4 are given in True Equator, Mean Equinox (TEME) format. TEME remains fixed to relative to the stars not to the Earth. It uses the Earths true equatorial plane at that moment in tim, but for the x-axis it uses the mean equinox which helps smooth out some of the short term wobbles in the Earths's rotation.

As TEME does not rotate with the Earth, but ground locations given in latitude and longitude do rotate with the Earth, for visibility and elevation calculations, both the satellite and the observer location need to be in the same rotating frame. For this reason TEME must be converted to Earth Centered Earth Fixed (ECEF) coordinate system [**?**].

1) TEME to ECEF: The conversion for TEME to ECEF is as follows ...

$$\mathbf{r}_{\text{ECEF}} = R_3(\theta_{\text{GMST}})\ \mathbf{r}_{\text{TEME}}, \tag{8}$$

Where:

- $R_3(\theta)$ is a rotation matrix describing a rotation about the Earth's Z–axis.
- θ_{GMST} is the Greenwich Mean Sidereal Time at the observation epoch.

This can be written explicitly as a rotation matrix as follows ...

$$R_3(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{9}$$

Meaning ...

$$\begin{bmatrix} x_{\text{ECEF}} \\ y_{\text{ECEF}} \\ z_{\text{ECEF}} \end{bmatrix} = \begin{bmatrix} \cos\theta_{\text{GMST}} & -\sin\theta_{\text{GMST}} & 0 \\ \sin\theta_{\text{GMST}} & \cos\theta_{\text{GMST}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{\text{TEME}} \\ y_{\text{TEME}} \\ z_{\text{TEME}} \end{bmatrix} \tag{10}$$

2) Latitude Longitude to ECEF:

Given:

- Latitude: ϕ (radians)
- Longitude: λ (radians)
- Altitude: h (meters above the WGS–84 reference ellipsoid)

WGS–84 Ellipsoid Constants:

- $a = 6378137.0$ m (equatorial radius)
- $f = \dfrac{1}{298.257223563}$ (flattening)
- $e^2 = 2f - f^2$ (first eccentricity squared)

The first step is to find the distance from Earth's center to the surface at a given latitude ...

$$N(\phi) = \frac{a}{\sqrt{1 - e^2 \sin^2\phi}}. \tag{11}$$

Knowing this means it is now possible to compute the ECEF coordinate.

$$X = (N(\phi) + h)\cos\phi\cos\lambda \tag{12}$$

$$Y = (N(\phi) + h)\cos\phi\sin\lambda \tag{13}$$

$$Z = \big(N(\phi)(1 - e^2) + h\big)\sin\phi \tag{14}$$

D. Visibility Logic

Once both the ground observer and the satellites propagated position are represented in ECEF the first filter check for visibility can be applied. This is based on computing the satellite's elevation angle relative to the observers horizon.

Given the satellite position,

$$r_{\text{satellite}}^{\text{ECEF}} = (s_x, s_y, s_z), \tag{15}$$

and the observer position,

$$r_{\text{observer}}^{\text{ECEF}} = (g_x, g_y, g_z), \tag{16}$$

the line of sight vector (LOS) can be represented as,

$$d = r_{\text{satellite}}^{\text{ECEF}} - r_{\text{observer}}^{\text{ECEF}} = (d_x, d_y, d_z), \tag{17}$$

Using the observer latitude and longitude, the LOS can be converted into local horizon coordinates. This can be done following (**??**) [**?**]

$$\begin{aligned} e &= -\sin\lambda\, d_x + \cos\lambda\, d_y, \\ n &= -\cos\lambda \sin\phi\, d_x - \sin\lambda \sin\phi\, d_y + \cos\phi\, d_z, \\ u &= \cos\lambda \cos\phi\, d_x + \sin\lambda \cos\phi\, d_y + \sin\phi\, d_z. \end{aligned} \tag{18}$$

Using this, the elevation angle can then be computed as the angle between the LOS vector and the local horizon [**?**].

$$\theta_{\text{el}} = \arctan\left(\frac{u}{\sqrt{e^2 + n^2}}\right) \tag{19}$$

Then, to filter out any satellites which are too close to the horizon or below the horizon, a mask angle of 10° is applied.

$$\text{el} \geq \theta_{\text{mask}}, \qquad \theta_{\text{mask}} = 10^\circ \tag{20}$$

E. Coverage Loop

F. TLE offline Fallback

G. Visualisations

V. Results

The three core metrics which will be compared are coverage availability (as a percentage of time), capacity throughput (Mbps) and latency (ms).

Coverage availability determines how often the user has service, this metric is based on visibility of satellites and a high level model of the beamwidth of a given satellite. This will be compared to urban and rural 5G networks. Throughput is measured in Mbps, this is averaged over a period of time and will also be compared to urban and rural 5G throughput.

Number of satellites visible over time

Plots (elevation vs. time, count of visible satellites, etc.)

Sample day simulation

Visualisations

Coverage heatmaps

Capacity throughput estimates

(a) 5G coverage in Ireland – 48 [**?**]

(b) 5G coverage in Ireland – Eir [**?**]

(c) 5G coverage in Ireland – Three [**?**]

(d) 5G coverage in Ireland – Vodafone [**?**]

Figure 5: Network operator 5G coverage maps

A. Throughput

B. latency

VI. Conclusions

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